Automation of a Water Bottling Process Using a Virtual PLC

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Abstract

This project details the design, modeling, and implementation of an advanced automated system for a water bottling process, utilizing a Programmable Logic Controller (PLC) emulated in a virtual environment using TIA Portal. The primary objective consists of the complete integration of the technological workflow, starting from sensory data acquisition and water pumping from the main reservoir, up to precise dosing and the evacuation of containers on a conveyor belt.

The system implements a hybrid control logic, blending the sequential control required by finite state machines with an intelligent component based on adaptive decision-making logic and hysteresis. The energy management and predictive maintenance module calculates real-time key performance indicators (KPIs) such as Specific Energy Consumption (SEC) and the carbon footprint.

The results obtained through the SCADA/HMI system demonstrate the algorithm's efficiency under continuous operating conditions, the stability of the transient regime, and the system's ability to correctly manage fault states (Emergency Stop), validating the solution as a viable "Digital Twin" for modern industrial applications.

Contents

1 Introduction

In the context of the current industrial revolution, known as Industry 4.0 [8], the automation of production processes is no longer limited to the simple actuation of mechanical elements, but involves the massive integration of information technologies to optimize workflows. The food and beverage industry, particularly water bottling lines, requires a high degree of hygiene, speed, and precision—objectives that can be achieved exclusively through the use of Programmable Logic Controllers (PLCs) [7].

This project proposes the development and virtual validation of an integrated measurement, control, and supervision system for a bottling station. The motivation for choosing this topic stems from the need to modernize classical technological lines by adding an "intelligence" layer, capable not only of controlling the filling process but also of monitoring energy costs and the ecological footprint in real time.

The project execution report covers all the fundamental design stages imposed by automation standards: from the mathematical modeling of the process and the selection of the optimal control strategy [4], to the implementation of a Human-Machine Interface (HMI) and a predictive diagnostics component [9].

2 Theoretical Background

To justify the design decisions of this automated system, a description of the fundamental concepts controlling contemporary industrial automation is necessary.

2.1 Architecture of Industrial Control Systems (PLC)

Programmable Logic Controllers (PLCs) represent the core architecture of modern factories [7]. Their operation is based on a continuous cycle (scan cycle) consisting of three major steps: reading the status of physical inputs (PII), sequentially executing instructions from the user memory, and writing the results to the output image memory (PIQ). In time-critical real-time applications, such as the millimetric positioning of bottles on the conveyor, variations in the cycle time (jitter) can introduce unacceptable errors. For this reason, modern control theory dictates the use of time interrupts (Cyclic Interrupts), which guarantee the execution of the algorithm at a strictly deterministic interval [1].

2.2 The Concept of Digital Twin and Industry 4.0

The digitalization of production processes is the central pillar of the Industry 4.0 paradigm [8]. Creating a *Digital Twin* involves developing a faithful virtual replica of a physical process [10]. Within this project, the TIA Portal software is used to emulate the physical behavior of water, motors, and sensors, allowing the validation of the control logic in a completely safe environment. This approach drastically reduces commissioning time and the risk of damaging real equipment.

2.3 Cyber-Physical Systems and Predictive Maintenance

The transition from reactive maintenance (replacing defective components) to predictive maintenance represents a defining characteristic of Cyber-Physical Systems (CPS) [9]. By mathematically integrating operational parameters (such as the electrical power consumed by motors over a duration Δt), the system can estimate the degradation state of the equipment. Modern algorithms use these data to issue warnings before a catastrophic failure occurs, thereby minimizing production downtime.

3 Definition of the Analyzed Process

This chapter describes the operational workflow, the variables involved, and the technical constraints that control the operation of the bottling system.

3.1 Technological Description

The automated process aims to manage the transfer of liquid from a main storage reservoir to a dosing unit and, subsequently, the precise bottling of water into containers that are transported through an assembly line. The system architecture comprises three interdependent subsystems [3]:

- **Storage and Supply System (TK-01):** Acts as the primary water source. It is equipped with an auto-refill algorithm (logical hysteresis) to maintain the optimal level between the limits of 20% and 95%.
- **Transfer and Dosing System (P-01 and V-01):** A hydraulic pump (P-01) extracts fluid from TK-01 and transfers it to the intermediate tank of the filling valve (V-01). The filling of the bottle is performed gravitationally.
- **Transport and Bottling System:** A conveyor belt driven by an electric motor moves the bottles. It must be stopped with high precision under the axis of the dosing valve upon receiving confirmation from the optical sensor.

3.2 Input and Output Variables

To interface the process with the central processing unit, digital and analog variables have been defined in accordance with industrial instrumentation standards [2]. These are detailed in Table 1.

3.3 Operating and Safety Constraints

To guarantee equipment safety and prevent raw material losses, the system imposes a series of strict logical interlocks [4]:

- Pump P-01 is software-interlocked if the level in reservoir TK-01 drops below the critical threshold of 5% (cavitation protection).

Table 1: List of primary process variables

Type	Tag Name	Technical Description	Range of Variation
DI	Senzor_RecipiPrezent	Optical detection of bottle under valve	TRUE / FALSE
DI	Senzor_RecipiIesire	Confirmation of bottle path completion	TRUE / FALSE
AI	Nivel_Rezervor	Current water level in TK-01	0.0 ... 100.0 %
DQ	Motor_Banda_Pornit	Conveyor contactor relay command	TRUE / FALSE
DQ	Ventil_Deschis	Dosing solenoid valve command	TRUE / FALSE
AQ	Comanda_Viteza_Pompa	Speed reference for pump P-01	0.0 ... 100.0 %

- The dosing solenoid valve (V-01) can open exclusively if the conveyor belt reports zero speed and the sensor confirms the presence of the container.
- Upon triggering a STOP or an Emergency Stop (EMG STOP), the process is "frozen". The bottle position and the dosed quantity are retained in Data Blocks (DBs), allowing the resumption of the process from the interruption state without generating defects.

3.4 Imposed Performance Requirements

The technological performance metrics targeted by this project are:

1. **Positioning Accuracy:** Deviation lower than 1% of the total conveyor travel.
2. **System Responsiveness:** Response time to external excitations of less than 100 ms.
3. **Hydraulic Stability:** Maximum oscillations of $\pm 2\%$ in the dosing tank during continuous operation.

4 Measurement and Acquisition System

This chapter details the elements necessary for converting the physical quantities of the bottling process into digital data processable by the PLC.

4.1 Identification of Quantities and Sensor Selection

Based on the control requirements formulated previously, industrial sensors with specific characteristics for food-grade applications were selected [2]:

- **Ultrasonic Level Sensor (for TK-01):** The operating principle is based on measuring the Time of Flight of a sound wave emitted toward the water surface. The output is a unified 4-20 mA signal, providing high immunity to electromagnetic noise over long cable runs.
- **Incremental Rotary Encoder (for the conveyor):** Attached to the shaft of the electric motor, this sensor transforms continuous mechanical displacement into digital pulses, allowing the PLC to compute a virtual coordinate of the bottle (position 0-1000).

- **Retro-reflective Photoelectric Sensor (for the valve):** Fixed aligned with the filling axis (coordinate $X = 541$), its role is to confirm the mechanical stop of the container before valve actuation.

4.2 Measurement Chain Diagram and Signal Conditioning

The data acquisition architecture follows the standard industrial workflow:

Physical Process → Sensor → Signal Conditioning (4-20mA) → A/D Conversion → Digital Filtering → Contro

Analog signals are acquired by the input modules of the S7-1500 unit [1]. The conversion of the raw value from the Analog/Digital converter (N_{PLC}) into the actual physical quantity (Level, L) is performed via a linear scaling algorithm:

$$L[\%] = \frac{N_{PLC} - N_{min}}{N_{max} - N_{min}} \times (L_{max} - L_{min}) + L_{min} \tag{1}$$

where, according to Siemens specifications [1], the nominal unipolar digitization interval is $N_{PLC} \in [0, 27648]$.

5 Measurement Error Analysis

Evaluating the accuracy of the measurement chain represents a critical stage, guaranteeing that the data processed by the controller accurately reflect the physical reality.

5.1 Uncertainty Calculation and Sampling Errors

According to the metrology of acquisition systems [2], the total level measurement error is influenced by the accuracy of the field sensor and the resolution of the PLC acquisition module.

1. **Quantization Error (A/D):** Due to the high-performance 16-bit module, the system resolution is extremely high. The deviation introduced is $\frac{100\%}{27648} \approx 0.0036\%$, a negligible value.

2. **Sensor Error (u_{sensor}):** A standard industrial ultrasonic sensor presents a manufacturer-declared uncertainty of $\pm 0.5\%$.

3. **Estimated Total Uncertainty (u_{total}):** Considering statistically uncorrelated variables, the combined standard uncertainty is calculated using the root sum of squares method:

$$u_{total} = \sqrt{u_{sensor}^2 + u_{ADC}^2} = \sqrt{(0.5)^2 + (0.0036)^2} \approx 0.50001\% \tag{2}$$

   The result demonstrates that the dominant error belongs exclusively to the sensor, with digital processing being transparent.

4. **Temporal Sampling Error:** The `OB35` block executes the reading of inputs at exactly 100 ms. This period generates an acceptable delay, translating into a negligible deviation of the bottle position at nominal conveyor speeds.

6 Communication and Supervision Architecture

This chapter evaluates the communication structure of the system and the integration of visualization modules (SCADA).

6.1 Structure Type and Communication Protocol

The system utilizes a **centralized structure**, based on a powerful primary processing unit (SIMATIC S7-1500) that concentrates the computation of PID algorithms and logical decisions [1]. Communication between the PLC and the HMI supervision system is realized, at a conceptual level, via the Industrial Ethernet/PROFINET protocol. This architecture provides a sufficiently high bandwidth for real-time transfer of diagnostics data (KPIs).

6.2 Supervision Integration (HMI / SCADA)

Process supervision is materialized through a WinCC Runtime Advanced application, which runs as a local SCADA system. Its role is twofold:

- It provides an **interactive graphical representation** of the machine state (animations showing bottle displacement, tank level indicators, valve signaling).
- It hosts an energy management **dashboard**, a fundamental requirement of the Digital Twin concept [10], allowing the graphical visualization of Overall Equipment Effectiveness (OEE) and consumption metrics.

The refresh rate (Acquisition Cycle) of the visual variables was set to Cyclic Continuous at 100 ms intervals, ensuring real-time fluency of animations on the operator's screen.

7 Control System Design

Designing the control logic involves establishing mathematical models and selecting optimal strategies to satisfy the strict performance criteria stated in Chapter 3.

7.1 Modeling of the Analyzed Process

The foundation of any reliable automated system is its mathematical modeling [4]. In this project, level control in the dosing tank is a critical subsystem.

The hydraulic dynamics of the V-01 tank can be approximated by the mass continuity equation: the tank cross-section multiplied by the level variation equals the difference between the inlet flow (pump P-01) and the outlet flow (valve). Applying the Laplace transform, the transfer function linking the level variation $H(s)$ to the pump flow $Q_{in}(s)$ takes the form of a first-order integrator system:

$$G(s) = \frac{H(s)}{Q_{in}(s)} = \frac{K}{T \cdot s + 1} \tag{3}$$

where K is the static gain (dependent on tank geometry) and T is the process time constant. The presence of this pure integrator behavior justifies the need for proportional control action, as a simple On-Off switch command would cause unamortized oscillations.

7.2 Selection and Justification of the Control Strategy

To manage process complexity, a **hybrid approach** was selected [3]:

1. **Sequential Control (Finite State Machine):** Intended exclusively for the conveyor belt. The transition from one state to another (Move → Detect → Stop → Fill → Reset) takes place based on strict logical conditions. This type of control is indispensable for the predictable cyclicity of the bottling process.

2. **Hysteresis Control:** Utilized for the main reservoir (TK-01). Maintaining the level between 20% and 95% is a standard industrial practice [4]. The major advantage is the physical protection of the pump against intermittent operating regimes (premature wear of contactors).

3. **Proportional Linear Control:** Used for commanding pump P-01, whose flow rate is dynamically calculated based on the system error $e(t)$ within the secondary tank.

8 Intelligent Component

In accordance with the design requirements, this chapter details the integration of concepts that expand beyond classical logic. An **Adaptive and Predictive Consumption Diagnostics System** has been developed.

8.1 Adaptive Resource Decision Logic

The intelligent system integrated into the PLC acts as an autonomous supervisor. The control logic for the flow rate of pump P-01 does not send a fixed command, but rather calculates the instantaneous error $e(t)$ between the desired level reference $r(t)$ and the actual value read by the sensor $y(t)$ [6]:

$$e(t) = r(t) - y(t) \tag{4}$$

By permanently adapting the pump speed through multiplying this error by a weighting factor, the system achieves dynamic balancing, supplying exactly the amount of water extracted by the filling valve without generating electrical energy overconsumption.

8.2 Predictive Performance Monitoring (KPIs)

The "intelligent" pillar of the project consists of its capacity to evaluate energy and operational efficiency, specific to Cyber-Physical implementations [5]. The PLC was programmed to calculate the following aspects in real time:

- **SEC (Specific Energy Consumption):** The aggregated electrical energy (in kWh) consumed across the entire line, divided by the number of bottles produced. It represents the energy cost per finished product.

- **Ecological Carbon Footprint:** Through an integrated scaling algorithm, the equivalent in kg of CO_2 generated for the consumed energy is calculated.

- **Predictive Analytics:** The system strictly monitors and integrates the operating time (in hours) of the motors and pumps. These counters allow the technical team to initiate preventive maintenance work based on calculated wear [9], rather than arbitrary time intervals.

This intelligent component transforms the project from a simple sequential automaton into a financial and technological decision-support tool.

9 Implementation Schemes

9.1 System Block Diagram

The functional architecture follows the standard process control paradigm. Field sensors collect raw information and convert it into electrical signals. These signals are digitized and filtered by the SIMATIC S7-1500 PLC. This is where the programmed logic (state machine and hysteresis) intervenes, deciding corrective actions that are sent back to the field as commands for electrical actuators (motors, valves). In the upper layer, the HMI extracts aggregated parameters (KPIs) for display.

9.2 Virtual Electrical Schematic (I/O Mapping)

Due to the Digital Twin implementation being exclusively in a simulated environment (S7-PLCSIM), the physical electrical schematic for power and control is substituted by memory organization inside TIA Portal. Hardware connections were emulated by precisely allocating tags to input (%I) and output (%Q) addresses, as well as to global retentive data blocks, which save energy statistics during power downs.

10 Implemented Algorithm

This chapter exposes critical code sequences implemented in SCL (Structured Control Language), a high-level programming language (similar to Pascal) ideal for manipulating mathematical formulas in PLCs [7]. Execution takes place inside the cyclic interrupt block 'OB35' (100ms).

10.1 Maintenance and Energy Integration Algorithm

To calculate the energy in kWh needed for the intelligent diagnostics module, the instantaneous power of the motors is integrated over the sampling time step:

```
// Pump Motor P-01 Monitoring
IF "Date_Pompa".Motor.Pornit THEN
    // P = U * I * power_factor (simplified to a coeff. of 0.22 for kW)
    "Date_Pompa".Motor.Putere_kW := "Date_Pompa".Motor.Curent_A * 0.22;

    // Integration of consumed energy (P * dt in hours)
    "Date_Pompa".Motor.Energie_kWh := "Date_Pompa".Motor.Energie_kWh +
                                      ("Date_Pompa".Motor.Putere_kW *
                                         0.000002778);
```

```
    // Total runtime hours for wear diagnostics
    "Date_Pompa".Motor.TimpFunctionare_h := "Date_Pompa".Motor.
        TimpFunctionare_h + 0.000027778;
END_IF;
```

Listing 1: Cumulative energy and time calculation for predictive maintenance

10.2 Hydraulic Protection and Adaptive Control

The following code demonstrates equipment protection using hysteresis logic, followed by a Specific Energy Consumption (SEC) calculation equation, protected against division-by-zero errors that would otherwise force the PLC into STOP mode.

```
// Automatic tank replenishment (hysteresis 20-95%)
IF "Date_Pompa".Nivel_Rezervor < 20.0 THEN
    "Date_Pompa".Alimentare_Activa := TRUE;
ELSIF "Date_Pompa".Nivel_Rezervor > 95.0 THEN
    "Date_Pompa".Alimentare_Activa := FALSE;
END_IF;

// Specific Energy Consumption (SEC) Calculation
IF "Date_Secventa".Productie.RecipiUmplute > 0 THEN
    "Date_Secventa".SEC := ("Date_Secventa".Energie_Totala * 1000.0) /
                            INT_TO_REAL("Date_Secventa".Productie.
                               RecipiUmplute);
ELSE
    "Date_Secventa".SEC := 0.0; // Math Fault protection during
        initialization
END_IF;
```

Listing 2: Auto-refill with hysteresis and financial SEC calculation

11 Experimental Testing and Simulation

Experimental validation of the concept was performed using advanced simulation tools. The closed-loop behavior of the system was evaluated via the SCADA interface, confirming compliance with the initial specifications.

11.1 Nominal Operating Regime Analysis

Figure 1 illustrates the equipment monitoring interface during a normal cycle. Correct actuation of the proportional pump, liquid transfer to the dosing tank (maintained within optimal parameters), and precise stopping of the bottle at the designated coordinate are observed. The visual system signals the state of each actuator through color codes (green-active, red-stopped), facilitating ergonomic supervision by the operator.

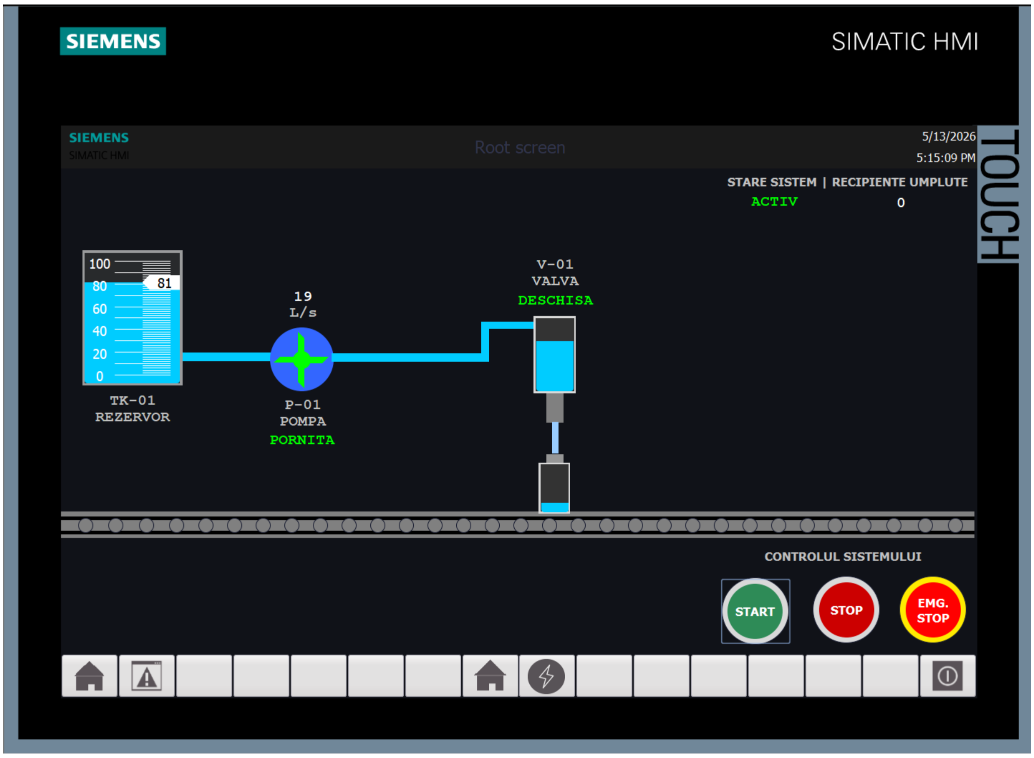

Figure 1: SCADA interface developed for real-time monitoring of the bottling process

11.2 Analysis of the Intelligent Maintenance System

The second direction of testing focuses on the energy management dashboard (Figure 2). The trend graph confirms the linear evolution of power during simultaneous motor operations. Decision-making intelligence algorithms process these raw data, returning precise decimal values for energy cost (in RON) and environmental impact (kg CO_2). The results demonstrate adequate sensitivity of the numerical simulation.

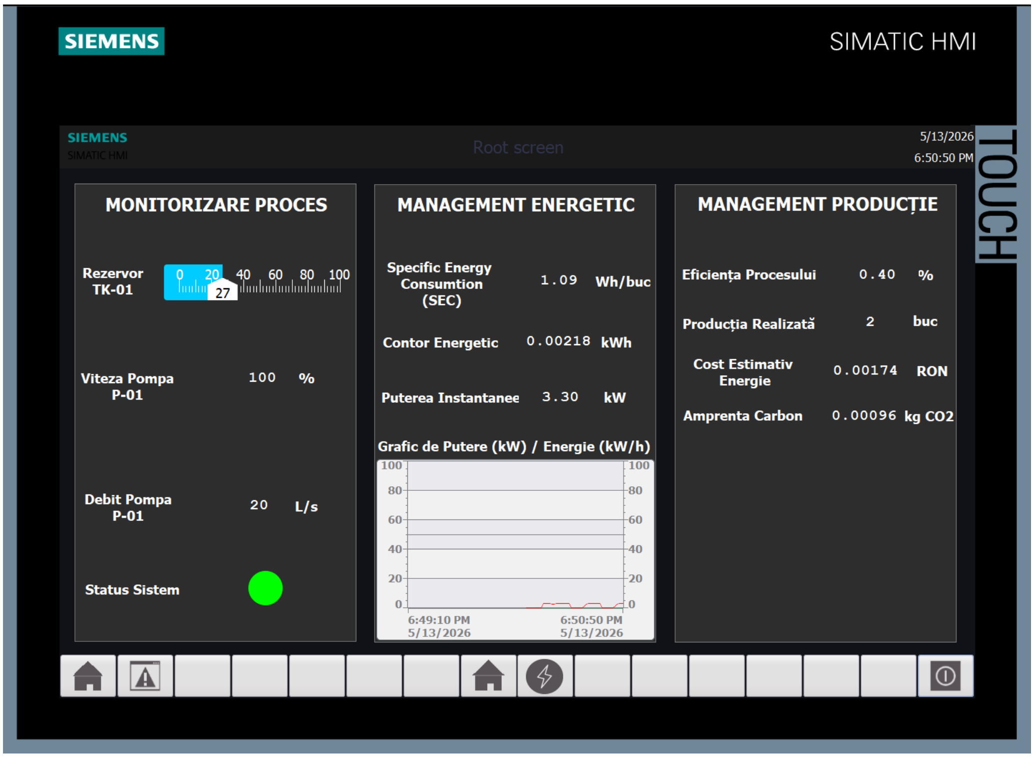

Figure 2: HMI Dashboard - Monitoring of performance indicators (SEC, OEE, CO2)

12 Critical Performance Analysis

This chapter evaluates system viability through four parameters essential to the modern industrial environment: robustness, scalabilitiy, cost, and integration.

12.1 Robustness and Reliability

The project distinguishes itself through remarkable robustness against logical perturbations. The use of a separate algorithm for managing crisis events (Emergency Stop) ensures the controlled stop of all moving elements. The true reliability innovation consists of retaining system states in retentive variables: upon power restoration, the system does not evacuate a partially filled bottle (which would constitute a defect), but continues filling with the remaining quantity instead. Implementing hysteresis radically decreases the mechanical switching frequency of pump P-01 relays [4].

12.2 Industrial Scalability

Code development followed object-oriented programming principles adapted for PLCs (the use of FB blocks instantiated multiple times, linked to dedicated DB data blocks). This architectural choice allows for **maximum scalability**: adding a second bottling line requires simply calling the same logical code block, without rewriting the core algorithms.

12.3 Estimated Cost and Economic Viability

Physical implementation of such a system requires a budget focused on Siemens hardware components, ultrasonic sensors, and motors driven by frequency converters. However, the investment is rapidly amortized thanks to the Intelligence and Energy Management component. By reducing peak consumption and planning repairs based on predictive data, the Total Cost of Ownership (TCO) decreases significantly [5].

12.4 Integration into Cyber-Physical Systems (CPS)

The proposed approach demonstrates that traditional industrial processes can be transformed into native components of the Industry 4.0 revolution [8]. The availability of energy and performance data (OEE) at the PLC unit level allows for their secure transmission, via OPC UA or MQTT, to cloud systems for complex Big Data analytics.

13 Conclusions

The present project successfully solved the problem of automating, optimizing, and monitoring a technological bottling flow, validating at a virtual level a solution ready for transfer onto physical equipment. By conjugating knowledge of instrumentation, adaptive control, and industrial informatics, a complete system that meets precision and productivity demands was created.

The main objectives achieved during the project elaboration are:

- **Metrological Development:** Quantities of interest were identified, and quantization errors of the proposed sensors were proven to be within negligible tolerances of $\approx 0.5\%$.
- **Control Stability:** By designing a hybrid control scheme (sequential on the conveyor, proportional on the dosing tank), hydraulic oscillation phenomena were eliminated, resulting in zero stationary positioning error at the filling point.
- **Intelligent Component Integration:** Beyond classical execution functions, the system proved its strategic utility by calculating economic impacts (Specific Energy Consumption - SEC) and maintenance cycle predictability in real time. This justifies its applicability within the "Smart Factory" concept.

In conclusion, the approach based on Digital Twin emulation offers major advantages during the development stage, as immediate visualization via the SCADA interface permits refined debugging of code and a substantial increase in the quality of the delivered technical solution.

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